

Pyridinium(Boron) Fluorophores as Versatile Photocatalysts

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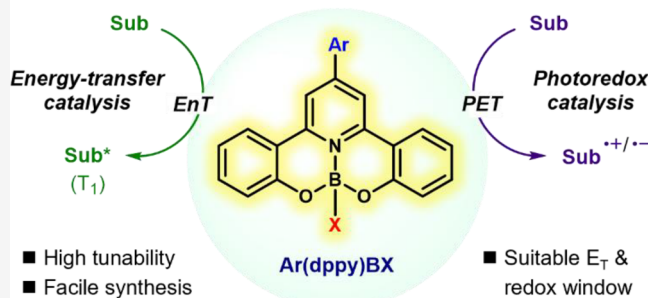
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ABSTRACT: Photoexcited molecules enable novel chemical transformations by exhibiting unique redox and photosensitizing properties. While various molecular photocatalysts have been developed, the ones that can facilitate a wide array of reactions remain scarce, and many existing photocatalysts lack the structural flexibility required for fine-tuning their properties. In this study, we introduce the pyridinium(boron) motif as a versatile fluorophore, forming a highly tunable family of organic photocatalysts with extensive applicability. The model compound Ph(dppy)BOMe showcases exceptional photoredox activity and energy transfer capabilities, rivaling those of widely used established photocatalysts. By strategically tailoring the substituents to suit specific catalytic reactions, we demonstrate that these catalysts can be optimized for divergent reactions requiring distinct redox characteristics. The straightforward modular synthesis and flexible tunability of pyridinium(boron) fluorophores position them as promising candidates for advanced photocatalytic applications.

Photocatalyst based on pyridinium(boron) fluorophore



1. INTRODUCTION

The photoexcitation of molecules results in the formation of excited states that exhibit properties distinct from their ground states, including unique redox and photosensitizing capabilities.^{1–18} By harnessing these properties, chemists can facilitate chemical transformations^{4,5,8,11,16,19,20} that are not possible within the constraints of the ground state potential energy surface. To date, a diverse array of fluorophore molecules, encompassing both organometallic and purely organic dyes, has been employed as photocatalysts (PCs) to enable photoredox and energy-transfer catalysis, leading to numerous novel photochemical transformations (Figure 1A).^{1,2,11,12,16,17,21} Each class of PC displays unique redox and photosensitization characteristics, making them suitable for a variety of specific reactions.

A review of several representative PCs utilized in synthetic studies indicates that, while organometallic compounds remain prevalent, there has been a significant increase in the application of organic PCs. Notably, 4CzIPN has emerged as a particularly effective fluorophore, with its utilization markedly increasing since its identification as a highly efficient PC in 2016.²² Mechanistically, 4CzIPN possesses an optimal excited state lifetime and favorable redox potential windows for both reductive quenching ($E_{red}^* = +1.35$ V, $E_{red} = -1.21$ V vs SCE) and oxidative quenching ($E_{ox}^* = -1.04$ V, $E_{ox} = +1.52$ V vs SCE), rendering it well-suited for interaction with a broad array of redox-active substrates.²³ Additionally, its moderate triplet energy (53 kcal/mol) enhances its versatility, rendering it an efficient triplet sensitizer for energy transfer catalysis.²² However, the core structure of this photocatalyst, which

consists of four donor units paired with a phthalonitrile acceptor moiety, presents limitations in tuning its redox properties. Specifically, modifying the redox characteristics necessitates the introduction of multiple altered donor units, while the acceptor component typically remains constant.²⁴ This constraint complicates the synthesis of modified Cz-PN photocatalysts and restricts the range of modifiable redox potentials. Therefore, there is an urgent need to identify novel fluorophore molecules that possess a simple and compact structure, with independently tunable donor and acceptor components. Such advancements could lead to the development of alternative PCs that complement or even surpass the capabilities of 4CzIPN across a broad spectrum of photocatalytic applications.

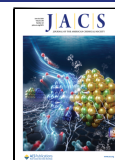
Our group has been exploring *N*-boryl pyridyl anion chemistry since 2017.²⁵ We discovered that the reaction of a diboron(4) compound, a pyridine molecule, and a methoxide anion yields the *N*-boryl pyridyl anion, which has been identified as a photoredox-active species that can be excited under visible light. This structure exhibits extreme reduction potentials, enabling a series of challenging single-electron reduction reactions (Figure 1B).^{19,26,27} This finding suggests that the *N*-borylpyridine structure could serve as a unique

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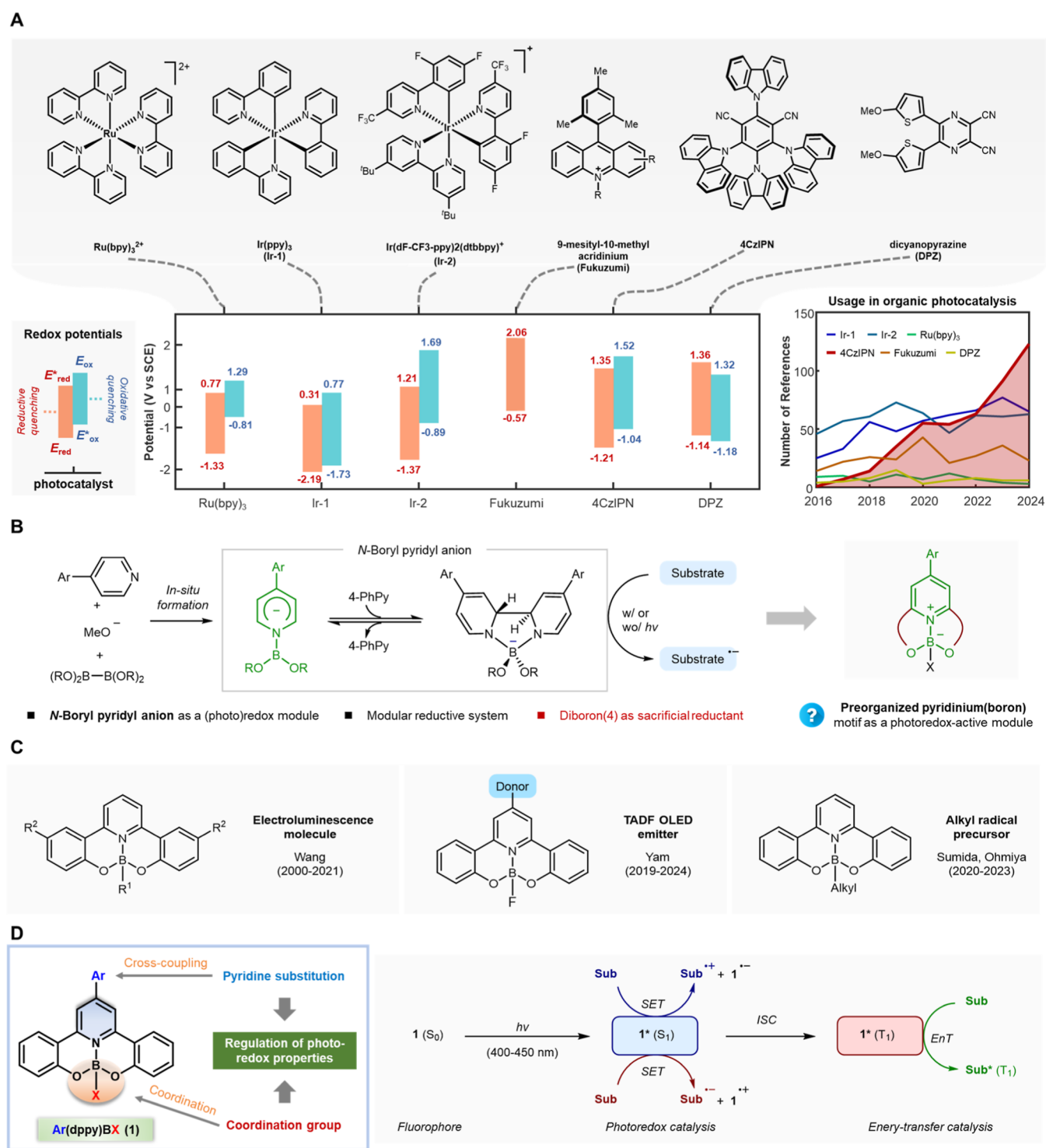


Figure 1. Catalysts involved in homogeneous photocatalysis. (A) Properties of representative molecular photocatalysts and their trend in synthetic use. (B) *N*-Boryl pyridyl anion chemistry and a postulated molecular redox system based on preorganized pyridinium(boron) motif. (C) Established applications of the pyridinium(boron) motif. (D) Photoredox and energy transfer catalysis enabled by pyridinium(boron) fluorophores (this work).

modular fluorophore with distinct photoredox properties. Building on this insight, we hypothesize that pyridinium(boron) motifs, wherein the *N*-borylpyridine structure is preorganized through covalent bonding, may serve as the fluorophore for a new type of photocatalyst.

A review of the literature reveals that the preorganized pyridinium(boron) structure, 2,2'-(pyridine-2,6-diyl)-

diphenolate boron, has been designed and investigated for applications in organic light-emitting materials and radical chemistry (Figure 1C). Wang, Zhang, and co-workers utilized 2,2'-(pyridine-2,6-diyl)diphenolate boron fluoride as an electroluminescent molecule in organic light-emitting devices (OLEDs)²⁷ and further explored the performance of various derivatives.^{28–30} The Yam group developed 4-donor-substi-

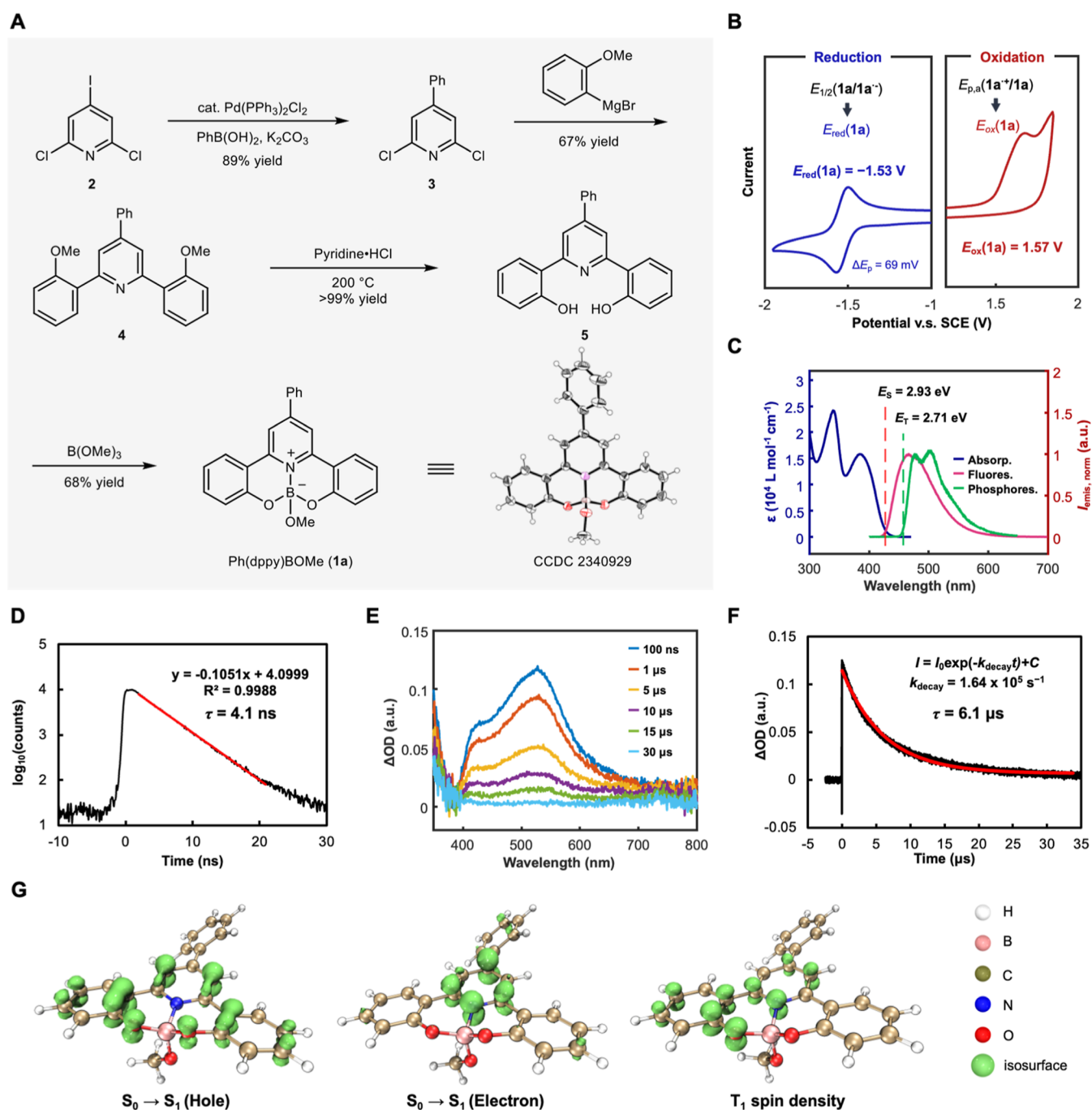
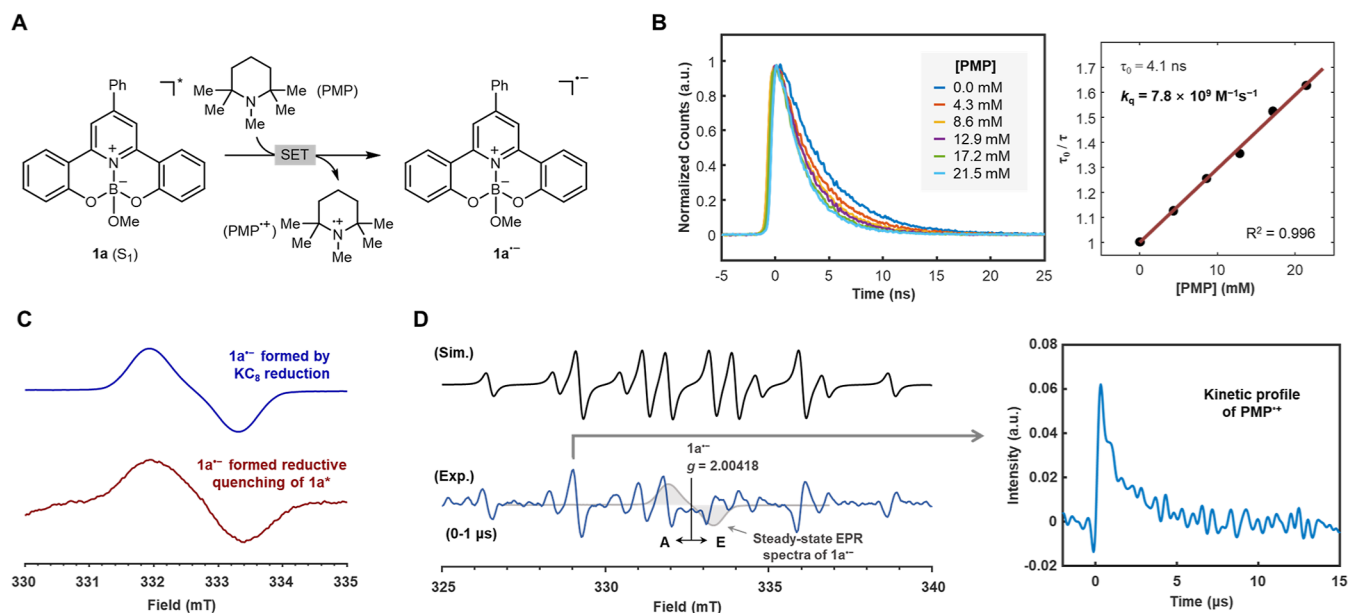


Figure 2. Synthesis and properties of Ph(dppy)BOMe (**1a**). (A) Synthetic route and XRD structure of **1a**. (B) Cyclic voltammogram of **1a** in MeCN. (C) UV-vis absorption and fluorescence spectra of **1a** in MeCN. (D) Time-correlated single photon counting spectrum of **1a** in MeCN (455 nm laser excitation, 490 nm probe). (E) TA spectra of **1a** in MeCN with 355 nm laser pulse excitation. (F) Kinetic trace for the decay of the excited transient species monitored at 528 nm. (G) Hole-electron analysis of $S_0 \rightarrow S_1$ excitation and the spin density of T_1 .

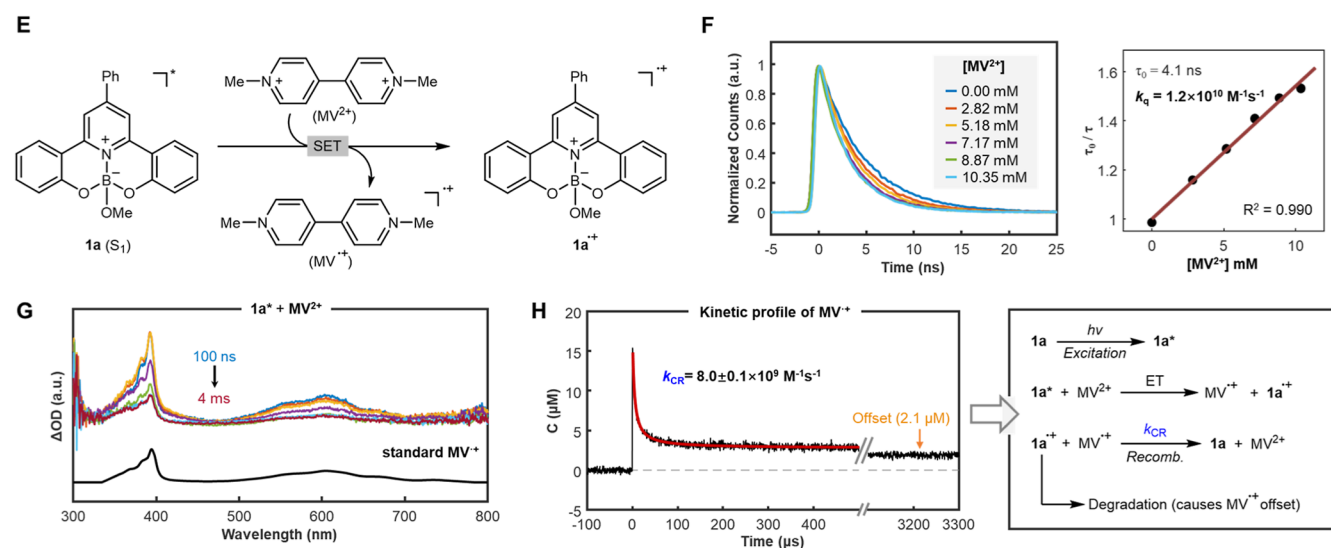
tuted 2,2'-(pyridine-2,6-diyl)diphenolate boron fluoride specifically for use as thermally activated delayed fluorescence (TADF) OLED emitters.^{31,32} Sumida, Ohmiya, and co-workers utilized 2,2'-(pyridine-2,6-diyl)diphenolate alkylboron as a reagent for generating alkyl radicals upon photoexcitation, which has been applied in several synthetic radical reactions.^{33–37} Interestingly, despite the characterization of the excited state properties of these pyridinium(boron) compounds in numerous studies,^{30,33} there remains a notable absence of research examining the potential of this fluorophore as a photocatalyst.

Inspired by the architecture of the existing 2,2'-(pyridine-2,6-diyl)diphenolate boron complex, we focus on the molecule Ar(dppy)BX (**1**). This molecule features a simple structure with independently tunable donor and acceptor components, allowing for modular synthesis. Therefore, we aim to investigate its potential as an efficient photocatalyst (Figure 1D). Herein, through a comprehensive investigation of the photophysical and photochemical properties of this molecule, we demonstrate that the Ar(dppy)BX fluorophore could effectively serve as an organic photocatalyst with both photoredox and energy transfer (EnT) capabilities. The

Reductive quenching of singlet excited state



Oxidative quenching of singlet excited state



Energy-transfer quenching of triplet excited state

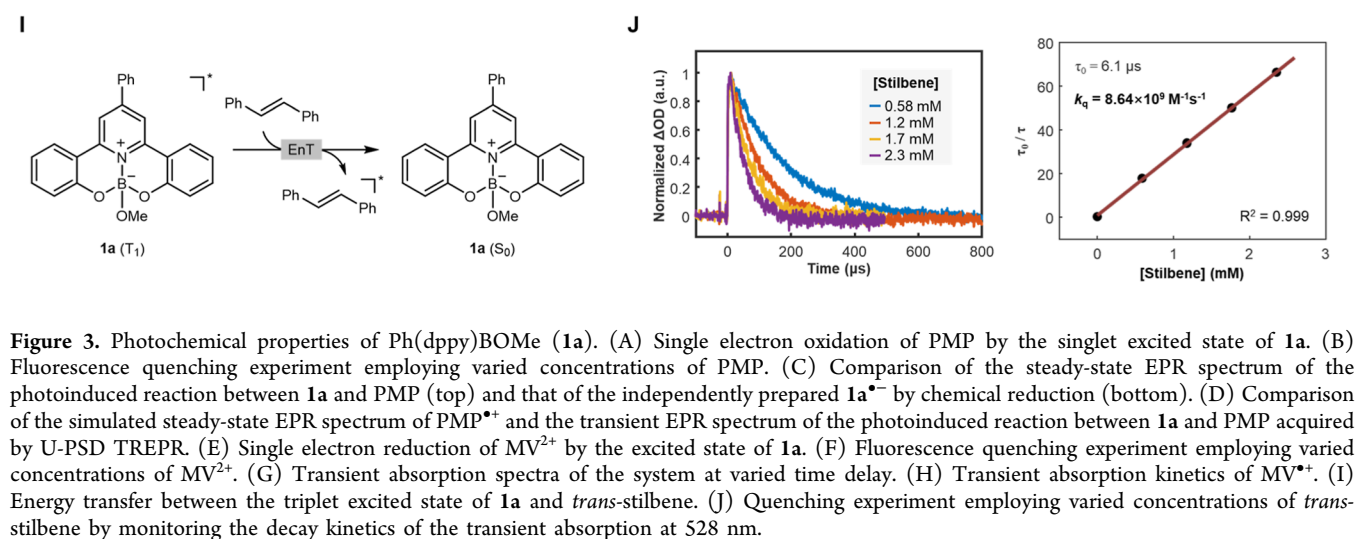


Figure 3. Photochemical properties of Ph(dppy)BOMe (**1a**). (A) Single electron oxidation of PMP by the singlet excited state of **1a**. (B) Fluorescence quenching experiment employing varied concentrations of PMP. (C) Comparison of the steady-state EPR spectrum of the photoinduced reaction between **1a** and PMP (top) and that of the independently prepared **1a^{•−}** by chemical reduction (bottom). (D) Comparison of the simulated steady-state EPR spectrum of PMP^{•+} and the transient EPR spectrum of the photoinduced reaction between **1a** and PMP acquired by U-PSD TREPR. (E) Single electron reduction of MV²⁺ by the excited state of **1a**. (F) Fluorescence quenching experiment employing varied concentrations of MV²⁺. (G) Transient absorption spectra of the system at varied time delay. (H) Transient absorption kinetics of MV^{•+}. (I) Energy transfer between the triplet excited state of **1a** and *trans*-stilbene. (J) Quenching experiment employing varied concentrations of *trans*-stilbene by monitoring the decay kinetics of the transient absorption at 528 nm.

versatility of the Ar(dppy)BX motif as a broad-spectrum photocatalyst in synthetic applications has been substantiated through its successful performance in various photocatalytic reactions. The ease of structural modification of this photocatalyst enables it to meet diverse requirements in different photocatalytic processes.

2. RESULTS AND DISCUSSION

2.1. Synthesis and Photophysical Properties of the Model Fluorophore Ph(dppy)BOMe

The model compound Ph(dppy)BOMe (**1a**) was synthesized in four steps with an overall yield of 41%, whose structure of was confirmed through single-crystal X-ray diffraction (XRD) analysis (Figure 2A). Cyclic voltammetry (CV) revealed a reversible reduction wave ($E_{1/2}(\mathbf{1a}/\mathbf{1a}^{\bullet-}) = -1.53$ V vs saturated calomel electrode, SCE, denoted as $E_{\text{red}}(\mathbf{1a})$) and an irreversible oxidation wave ($E_{\text{pa}}(\mathbf{1a}^{\bullet+}/\mathbf{1a}) = +1.57$ V vs SCE, denoted as $E_{\text{ox}}(\mathbf{1a})$) for **1a** (Figure 2B). These findings imply that $\mathbf{1a}^{\bullet-}$ acts as a moderately strong single-electron reductant, while $\mathbf{1a}^{\bullet+}$ demonstrates moderate oxidizing capability and may encounter degradation in actual catalytic reactions.

Photophysical studies of compound **1a** were conducted in acetonitrile (Figure 2C). UV-vis spectroscopy revealed a maximum absorption peak at 385 nm; fluorescence spectroscopy showed an emission peak at 466 nm; phosphorescence spectroscopy exhibited peaks at 477 and 502 nm. These results indicated that the singlet excited state energy (E_{S}) of **1a** is approximately 2.93 eV and the triplet energy (E_{T}) of **1a** is approximately 2.71 eV. These characteristics align with similar (dppy)B complexes reported in the literature,³² confirming the (dppy)B motif as an effective fluorophore. Application of the Rehm-Weller equation (see Figure S5 for corresponding Latimer diagram) indicated that **1a** possesses strong excited-state oxidizing ability ($E_{\text{red}}^* = +1.39$ V vs SCE) and moderate reducing capability ($E_{\text{ox}}^* = -1.35$ V vs SCE).

Compound **1a** showed a singlet state (S_1) lifetime of 4.1 ns in time-correlated single photon counting (TCSPC) experiments, which is sufficient to facilitate bimolecular reactions in solution (Figure 2D). Nanosecond transient absorption (TA) spectroscopy revealed a longer-lived species with an absorption maximum at 528 nm (Figure 2E) and a lifetime of 6.1 μs (Figure 2F). The lifetime decreased in aerated MeCN, suggesting that it corresponds to the triplet excited state of **1a** (Figure S7).

Time-dependent density functional theory (TD-DFT) calculations on **1a** were employed to gain a deeper understanding of its excited-state (Figure 2G). Hole-electron analysis³⁸ revealed that electron transfer from phenolate to pyridine subunit during $S_0 \rightarrow S_1$ excitation. For the T_1 state of **1a**, electron spin density analysis revealed that the spin density was distributed across both the pyridine and phenolate moieties. These results indicated that the **1a**'s excited states display partial charge separation (consistent in both S_1 and T_1), resembling a system with an electron-rich pyridylboron radical and an electron-deficient phenolate boron radical cation.

The electrochemical and photophysical properties of **1a** facilitate a comparative analysis of the Ar(dppy)BX fluorophore and 4CzIPN. In the context of reductive quenching, **1a** ($E_{\text{red}}^* = +1.39$ V and $E_{\text{red}} = -1.53$ V vs SCE) demonstrates a similar yet broader redox window compared to 4CzIPN ($E_{\text{red}}^* = +1.35$ V and $E_{\text{red}} = -1.21$ V vs SCE). Under oxidative

quenching conditions, **1a** ($E_{\text{ox}}^* = -1.35$ V and $E_{\text{ox}} = +1.57$ V vs SCE) also exhibits a border redox window than that of 4CzIPN ($E_{\text{ox}}^* = -1.04$ V and $E_{\text{ox}} = +1.52$ V vs SCE). Additionally, the E_{T} of **1a** (62 kcal/mol) significantly exceeds that of 4CzIPN (58 kcal/mol).³⁹ Although the excited state lifetimes of **1a** ($\tau_{\text{S}} = 4.1$ ns, $\tau_{\text{T}} = 6.1$ μs) are shorter than those of 4CzIPN ($\tau = 18$ ns, $\tau_{\text{T}} = 6.1$ μs), they remain adequate for facilitating bimolecular interactions with substrates. Reverse intersystem crossing behavior was not observed for **1a**, due to its greater singlet-triplet energy gap (ΔE_{ST}). Based on these characteristics, it can be inferred that **1a** possesses photocatalytic performance that is not only comparable to but may also surpass that of 4CzIPN, particularly in reductive quenching and energy transfer catalysis.

3. PHOTOCHEMICAL PROPERTIES OF PH(DPPY)BOME

To correlate the photophysical properties of **1a** with its chemical reactivity, interactions between excited **1a** ($\mathbf{1a}^*$) and quencher molecules were investigated.

Pempidine (PMP) was used to probe the redox properties of $\mathbf{1a}^*$. The singlet excited state (S_1) of **1a** exhibited sufficient reduction potential ($E_{\text{red}}^* = +1.39$ V vs SCE) to oxidize PMP ($E_{\text{ox}} = +0.73$ V vs SCE⁴⁰), forming PMP radical cation ($\text{PMP}^{\bullet+}$) and $\mathbf{1a}^{\bullet-}$ (Figure 3A). This photoinduced electron transfer (PET) process allowed us to investigate the photoredox properties of the Ar(dppy)BX fluorophore.

Quenching experiments revealed a linear increase in the fluorescence decay rate with increasing PMP concentration, yielding a quenching rate constant k_{Q} of $7.8 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$ (Figure 3B). Further TA kinetic studies showed that the presence of PMP significantly reduced the formation of the T_1 state of **1a** (Figure S16). These results suggest that the S_1 state of Ar(dppy)BX has a highly efficient interaction with PMP and primarily governs the photoredox process, rather than the T_1 state.

The photoredox process was further validated by observing product radicals. Steady-state electron paramagnetic resonance (EPR) detected the radical anion $\mathbf{1a}^{\bullet-}$, formed through reductive quenching of excited $\mathbf{1a}^*$ by the amine. Its identity was confirmed by comparing the EPR spectrum with that of $\mathbf{1a}^{\bullet-}$ independently synthesized via chemical reduction of **1a** using KC_8 (Figure 3C). These observations align with the stability of $\mathbf{1a}^{\bullet-}$ suggested by the reversible reduction peak in CV diagram.

The direct observation of another product radical, $\text{PMP}^{\bullet+}$, provided compelling evidence for the PET process. Despite that the detection of transient $\text{PMP}^{\bullet+}$ by conventional EPR technique is not possible, we settled this challenge by employing our newly developed ultrawide single sideband phase-sensitive detection (U-PSD) time-resolved EPR technique⁴¹ to investigate the interaction between compound **1a** and PMP. The characteristic time-resolved EPR spectrum of $\text{PMP}^{\bullet+}$, observed with a delay time of 0–1 μs , showed a splitting pattern revealing hyperfine interactions involving ^{14}N and $\beta\text{-C-H}$ ($a_{\text{N}} = 2.04$ mT, $3a_{\text{H}} = 2.75$ mT), which are consistent with previously reported values (Figure 3D).⁴² The spectrum also exhibited polarization with an absorption-emission pattern centered near the g -value of $\mathbf{1a}^{\bullet-}$, suggesting the formation of a singlet radical-pair during the electron transfer process.⁴³ This observation indicates that the S_1 state of **1a** plays a predominant role in facilitating the PET process, corroborating our TCSPC results. By tracking the character-

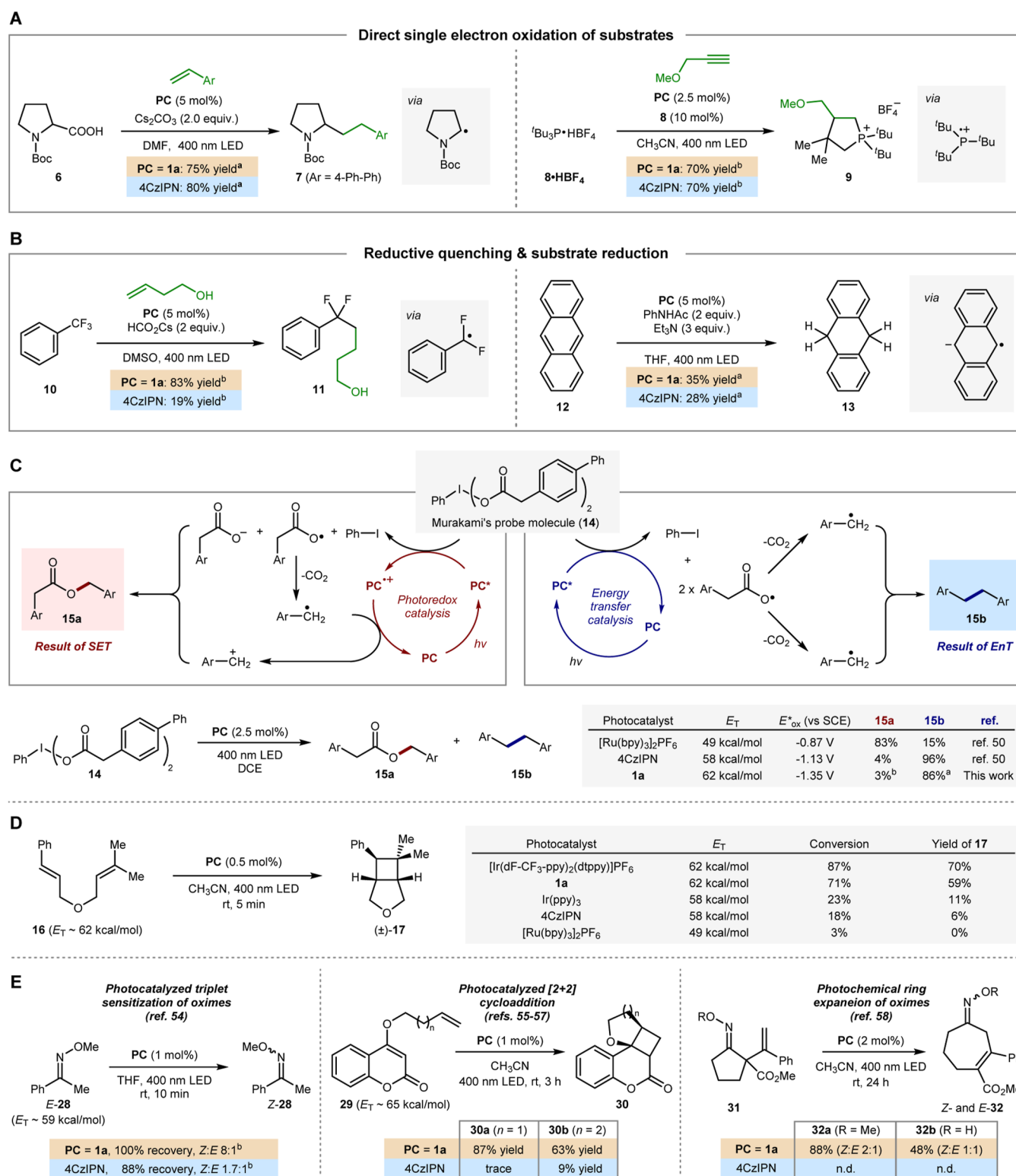


Figure 4. Evaluation of the photocatalytic performance of **1a**. (A) Direct SET oxidation reactions. (B) Reductive quenching reactions. (C) Probing the mode of action with Murakami's probe molecule. (D) Benchmarking the photosensitization reactivity in an intramolecular [2 + 2] cycloaddition. (E) Energy transfer reactions. ^aYields of isolated products are reported. ^bYields were determined by ¹H NMR analysis using 1,3,5-trimethoxybenzene as an internal standard.

istic peaks in the EPR spectra, we can observe the decay kinetics of PMP^{•+}, revealing a lifespan of approximately 7 μs (Figure 3D, right). Collectively, these findings provide a comprehensive understanding of the reductive quenching of **1a**^{*}, supported by extensive spectroscopic and kinetic data.

Methyl viologen bis(hexafluorophosphate) (MV²⁺) was employed as a probe to investigate the oxidative quenching process of **1a** (Figure 3E). Fluorescence quenching experiments revealed a linear increase in the observed decay rate as a function of MV²⁺ concentration, yielding a quenching rate

constant (k_q) of $1.2 \times 10^{10} \text{ M}^{-1}\text{s}^{-1}$ (Figure 3F). Nanosecond TA spectroscopy allowed for the clear observation of the resulting $\text{MV}^{\bullet+}$, which is consistent with previously reported spectrum (Figure 4G).^{44,45} Notably, although most generated $\text{MV}^{\bullet+}$ underwent immediate charge recombination (CR) with $\text{1a}^{\bullet+}$, a significant fraction of $\text{MV}^{\bullet+}$ persisted in the system 4 ms after the excitation pulse. Quantitative kinetic analysis of the $\text{MV}^{\bullet+}$ signal indicated that approximately 15% of the $\text{1a}^{\bullet+}$ species underwent degradation following a single excitation pulse (Figure S13), and the remaining population of $\text{1a}^{\bullet+}$ underwent CR with $\text{MV}^{\bullet+}$ with a k_{CR} of $8.0 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$.

Trans-stilbene ($E_T = 49 \text{ kcal/mol}$)⁴⁶ was employed to study the photosensitization properties of 1a^* (Figure 3I). TA kinetic studies showed a significant decrease in the lifetime of 1a^* with increasing [*trans*-stilbene], yielding a k_Q of $8.64 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$ (Figure 3J). Additionally, TCSPC experiments confirmed that the fluorescence lifetime of 1a remained unaffected by *trans*-stilbene, indicating that *trans*-stilbene does not interact with the S_1 state of 1a (Figure S9).

Overall, these experimental results demonstrate that compound 1a functions effectively as both a photoredox-active molecule (in both direction) and a triplet sensitizer. This dual capability enables it to facilitate either single-electron transfer or energy transfer with substrate molecules, similar to the behavior observed with 4CzIPN. Consequently, this characteristic positions 1a as a promising candidate for advanced photochemical applications.

4. APPLICATION OF AR(DPPY)BX AS PHOTOCATALYSTS IN SYNTHESIS

4.1. Assessment of the Photocatalytic Performance

4.1.1. Photoredox Catalysis. To evaluate the performance of photocatalyst 1a , we tested several photoinduced redox reactions. In direct SET oxidation reactions (Figure 4A), such as the decarboxylative alkylation of proline derivative 6^{47} and the formal [3 + 2] cycloaddition of $t\text{Bu}_3\text{P}$ with terminal alkynes, 1a exhibited yields comparable to the widely used photocatalyst 4CzIPN.⁴⁸ These reactions involve critical steps such as single-electron oxidation and radical generation, where 1a performed reliably.

In reductive quenching reactions (Figure 4B), 1a outperformed 4CzIPN in the defluorinative alkylation of benzonitrile (yield of 11 : 83% vs 19%) and a Birch-type reduction of anthracene. The defluorinative alkylation, which used cesium formate as the reductant, proceeded smoothly without requiring the HAT catalyst PhSH.⁴⁹ Similarly, the Birch-type reduction using Et_3N as the reductant achieved superior results with 1a .¹⁹ These findings underscore the competence of 1a in many advanced photoredox processes. However, we found that in carboxylation reaction of benzylic tertiary C–H bond with CO_2 ,^{50,51} 1a was less reactive than 4CzIPN, suggesting that catalysts with different structural characteristics exhibit distinct advantages in photocatalysis (see Supporting Information Section 7.5).

4.1.2. Energy-Transfer Catalysis. The potential of 1a as an energy transfer (EnT) catalyst was explored using probe molecule 14 , which undergoes a Hofer–Moest reaction or a Kolbe dimerization under single-electron transfer or energy transfer conditions respectively (Figure 4C).⁵² Previous studies indicated that photocatalysts with low E_T values initiated a photoredox process to yield product 15a , while those with higher E_T values facilitated a photosensitization process to give

product 15b . Thus, the chemoselectivity of this reaction serves as an indicator of the role of the photocatalyst: 1a predominantly yielded product 15b (86%) with minimal 15a (3%), confirming its efficacy as an EnT catalyst.

Given its high E_T of 62.5 kcal/mol, comparable to efficient triplet sensitizers like $[\text{Ir}(\text{dF-CF}_3\text{-ppy})_2(\text{dtbbpy})]\text{PF}_6$ and surpassing $\text{Ir}(\text{ppy})_3$, 4CzIPN, and $\text{Ru}(\text{bpy})_3^{2+}$, we hypothesized its potential in realizing more challenging EnT catalytic reactions. This was validated using an intramolecular [2 + 2] cycloaddition of substrate 16 , which has a high E_T (~62 kcal/mol) (Figure 4D).^{46,53} In this benchmark reaction, 1a outperformed catalysts with lower E_T values and showed comparable effectiveness to $[\text{Ir}(\text{dF-CF}_3\text{-ppy})_2(\text{dtbbpy})]\text{PF}_6$.

A broader survey demonstrates the utility of 1a as a highly efficient photocatalyst across a series of EnT reactions, where it outperforms benchmark organic photosensitizers (Figure 4E). For instance, in the photoisomerization of oxime 28 reported by the Rovis group using Ir complexes,⁵⁴ 1a outperformed 4CzIPN by driving the reaction to near completion within 10 min. Furthermore, in the [2 + 2] cycloaddition of 4-hydroxycoumarin derivatives 29 , which conventionally requires UV light (300 nm),⁵⁵ Ir catalysts (415 nm),⁵⁶ or specialized aminoboranes (390 nm),⁵⁷ 4CzIPN proved inactive, whereas 1a exhibited satisfactory efficiency under 400 nm irradiation. Similarly, in the photochemical ring expansion of oximes recently reported by Huang and co-workers, visible-light reactivity was previously exclusive to Ir complexes.⁵⁸ Under 400 nm irradiation, 1a successfully promoted this transformation, while 4CzIPN completely failed. These results establish 1a as a versatile organic EnT catalyst and a viable, metal-free alternative to established iridium-based systems in certain cases.

Based on these experimental evaluations, we propose that the pyridinium(boron)-based molecule 1a can serve as a versatile photocatalyst across a diverse range of photochemical reactions.

4.2. Photocatalyzed Switchable 1e/2e-Oxidation of Amines

The utility of photocatalyst 1a in synthesis was demonstrated by its application in the photocatalyzed switchable single electron/double electron oxidation of amines. The direct α -C–H functionalization of amines has garnered significant attention, particularly within the realm of photoredox catalysis. A breakthrough by the MacMillan group revealed that dialkyl anilines can undergo α -C–H arylation with electron-deficient benzonitriles using $\text{Ir}(\text{ppy})_3$ as a photocatalyst, yielding benzylic amines.⁵⁹ This process involves the 1e-oxidation of the amine substrate by the excited state of the photocatalyst, producing an aniline radical cation, which subsequently undergoes deprotonation at the α -position to generate an α -amino radical. This transformation has also been successfully achieved through an electrochemical endeavor invoking paired electrolysis⁶⁰ and via photoinduced reactions involving electron donor–acceptor (EDA) complexes.⁶¹ On the other hand, it has been demonstrated that the same substrates can undergo 2e-oxidation under both photocatalytic and electrochemical conditions, leading to the formation of iminium ions, which can ultimately yield α -cyanoamines as final products.

The control of 1e/2e-oxidation reaction pathway of the amine substrate represents a major challenge in this chemistry. The Luo group reported an electrochemical strategy to selectively control 1e/2-electron oxidation by employing

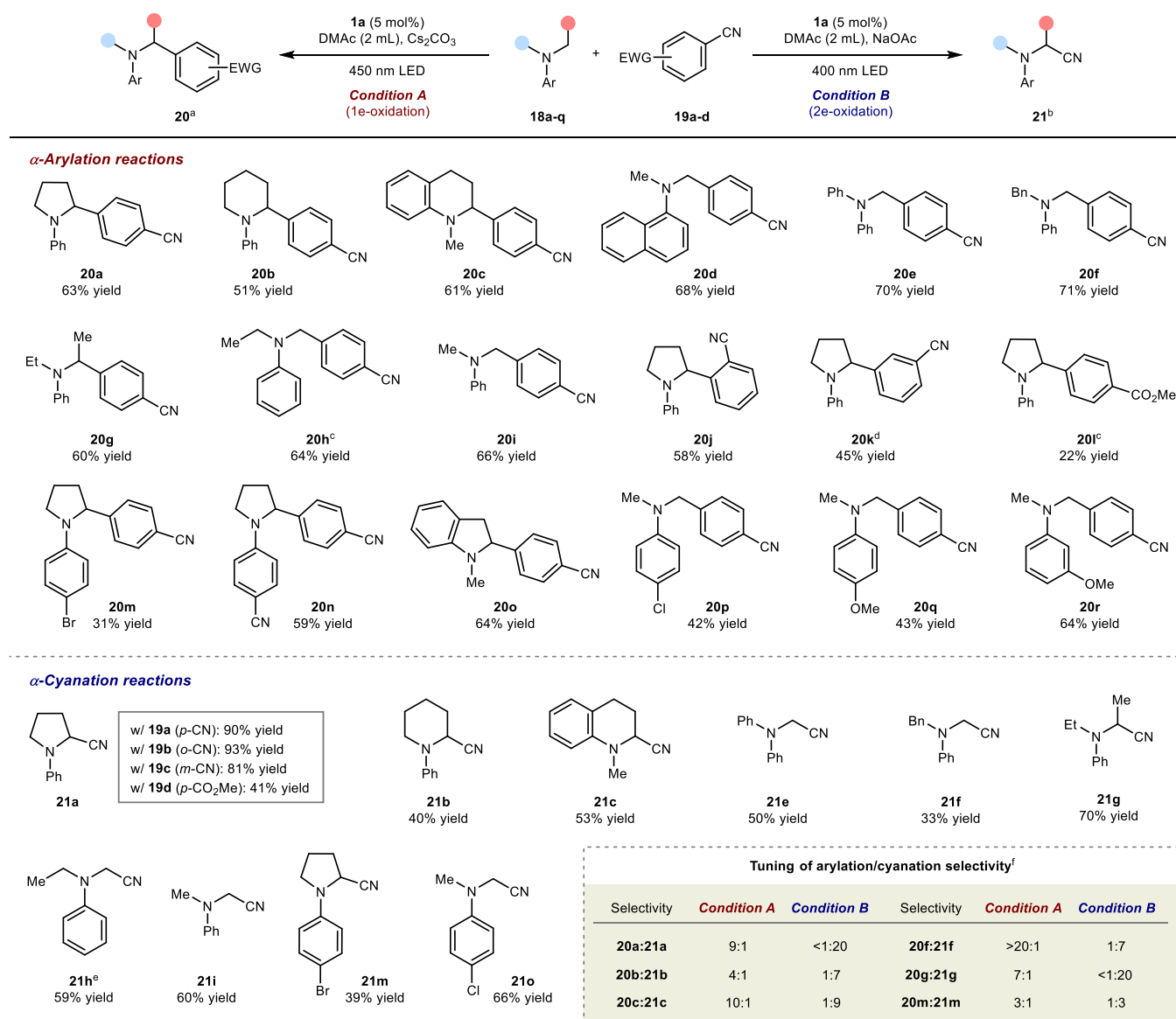


Figure 5. Substrate scope and the selectivity trend of α -arylation and α -cyanation reactions of amines catalyzed by Ph(dppy)BOME. ^aReaction conditions: **18** (1.5 equiv), **19** (0.5 mmol, 1.0 equiv), **1a** (5 mol %), Cs₂CO₃ (1.5 equiv) in 2 mL of DMAC (*N,N*-Dimethylacetamide), 20 W LED, Ar atmosphere, rt, 2 h. Yields of isolated products are reported. ^bReaction conditions: **18** (3.0 equiv), **19** (0.5 mmol, 1.0 equiv), **1a** (5 mol %), NaOAc (2.0 equiv) in 2 mL of DMAC, 20 W LED, Ar atmosphere, rt, 24 h. Yields of isolated products are reported. ^cSite Selectivity: CH₃:CH₂ = 7:1. ^dreaction time: 12 h. ^eSite selectivity: CH₃:CH₂ = 3:1. ^fTuning of arylation/cyanation selectivity were determined by isolated yield. Isolated yield without specifically labeled indicated that no other product was observed.

varying alternating current (AC) and direct current (DC) electrolysis, achieving moderate selectivity.⁶² However, to date, there have been no reports detailing the successful control of 1e/2e-oxidation pathways in photocatalytic systems.

We observed that, the choice of reaction conditions significantly influenced the 1e/2e-oxidation pathway of the aniline substrate **18** when utilizing **1a** as the photocatalyst. By varying these conditions, we successfully directed the reaction toward the formation of either the α -arylation product **20** or the α -cyanation product **21** (Figure 5). For the α -arylation of amine **18** with benzonitrile **19**, the optimal conditions were established as the use of a catalytic amount of **1a**, 450 nm LED irradiation, and Cs₂CO₃ as the base. This reaction exhibited a broad substrate scope, with *N*-phenyl pyrrolidine, piperidine, and tetrahydroquinoline reacting efficiently with *para*-dicyanobenzene (**19a**) to yield α -arylation products **20a–c** in good

yields. Additionally, acyclic amines participated effectively in the reaction, resulting in arylation products **20d–i**. Notably, the arylation occurred selectively at the less hindered α -positions of the amine alkyl substituents, exemplified by products **20f** and **20h**. Furthermore, ortho- (**19b**) and meta-dicyanobenzene (**19c**) were also compatible coupling partners, yielding products **20j** and **20k**, while methyl 4-cyanobenzoate (**19d**) showed diminished reactivity, resulting in product **20l**. Both electron-withdrawing and electron-donating substituents on the *N*-aryl ring of the substrate were well tolerated, affording moderate yields of products **20m–q**. Under the optimized conditions, the ratio of α -arylation to α -cyanation products generally exceeded 3:1, indicating a pronounced selectivity for the 1e-oxidation pathway.

Conversely, when we employed similar reaction conditions with **1a** as the photocatalyst, but substituted NaOAc as the

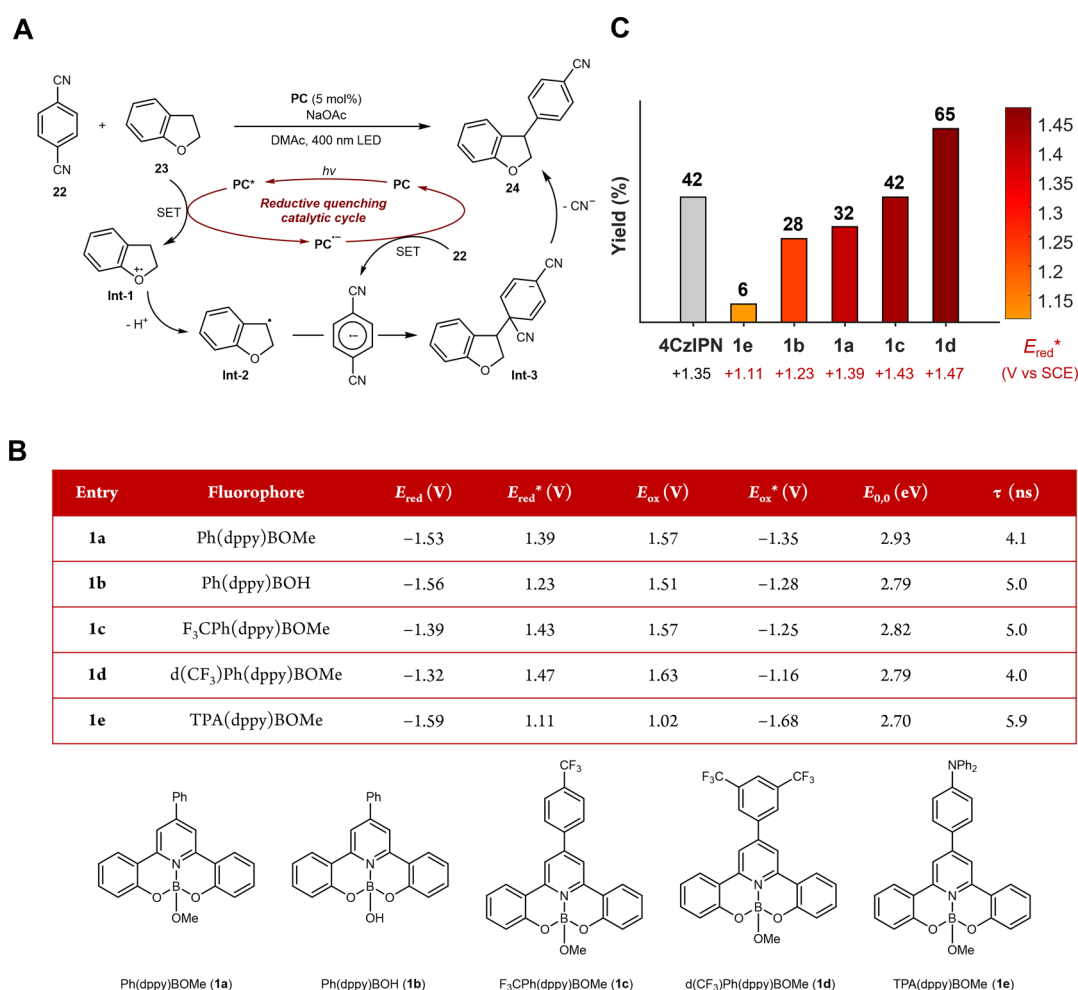


Figure 6. Structural tailoring of Ar(dppy)BX fluorophores for reductive quenching catalysis. (A) Model reaction: the C–H arylation of **23** with **22**. (B) Modifying the aryl group on pyridine. Redox potentials are referenced to SCE. (C) Evaluation of catalytic performance. Reaction conditions: **22** (0.5 mmol, 1.0 equiv), **23** (3.0 equiv), PC (5 mol %), NaOAc (2.0 equiv) in 2 mL of DMAC, 20 W LED, Ar atmosphere, rt, 24 h. Yields were determined by ¹H NMR analysis using 1,3,5-trimethoxybenzene as an internal standard.

base and utilized a 400 nm LED light source, we predominantly obtained α -cyanation products **21** from the same substrate set. The source of the cyano group exhibited a marginal effect on the reaction outcomes; specifically, the reaction of *N*-phenyl pyrrolidine with dicyanobenzenes **19a–c** produced good yields of α -cyanation product **21a**, whereas the less reactive methyl 4-cyanobenzoate yielded only moderate results. Both cyclic and acyclic anilines were compatible with the cyanation reaction, resulting in moderate yields (**21b–i**). Substituents on the *N*-aryl ring were also well tolerated (**21m** and **21o**). In all examined cases, the α -cyanation to α -arylation product ratio consistently exceeded 3:1, demonstrating a reversal in chemoselectivity that favors the 2e-oxidation pathway. While a comprehensive mechanistic elucidation of this switchable chemoselectivity is currently under pursuit, this protocol establishes a practical and tunable approach to divergent α -functionalized amines from a common set of starting materials.

4.3. Tailoring Ar(dppy)BX Catalysts for Specific Reactions

The modular architecture of the Ar(dppy)BX framework enables the precise alignment of a photocatalyst's photoredox properties with the energetic requirements of specific chemical transformations. By modifying the Ar-substituent on the pyridinium core or the X-group at the boron center, both

redox potentials and excited-state dynamics can be finely tuned. This structural versatility may provide a powerful toolbox for rational catalyst optimization across divergent catalytic cycles. Two representative photoredox catalytic reactions involving either reductive quenching and oxidative quenching processes were selected to demonstrate this feature.

4.4. Optimization for Reductive Quenching Catalysis

The efficacy of this tailoring strategy was first validated in a reductive quenching cycle (Figure 6A), specifically the C–H arylation of dihydrobenzofuran (**23**) with *p*-dicyanobenzene (**22**). In this oxidation-initiated process, catalytic efficiency is primarily governed by the excited-state reduction potential (E_{red}^*) of the catalyst. Through aryl group modification, a series of photocatalysts with distinct photoredox properties were obtained (Figure 6B). Compared to **1a**, the introduction of electron-withdrawing groups (EWGs) to the pyridinium core systematically enhanced excited-state oxidation potential while diminishing its excited-state and radical anion reduction capabilities. Specifically, compounds **1c** (Ar = 4-trifluoromethylphenyl) and **1d** (Ar = 3,5-bis(trifluoromethyl)phenyl) exhibited sequential increases in E_{red}^* and decreases in reduction ability. Conversely, **1e**, bearing a 4-diphenylamino-phenyl substituent, showed reduced oxidation and enhanced reduction abilities, highlighting the role of EWGs. This

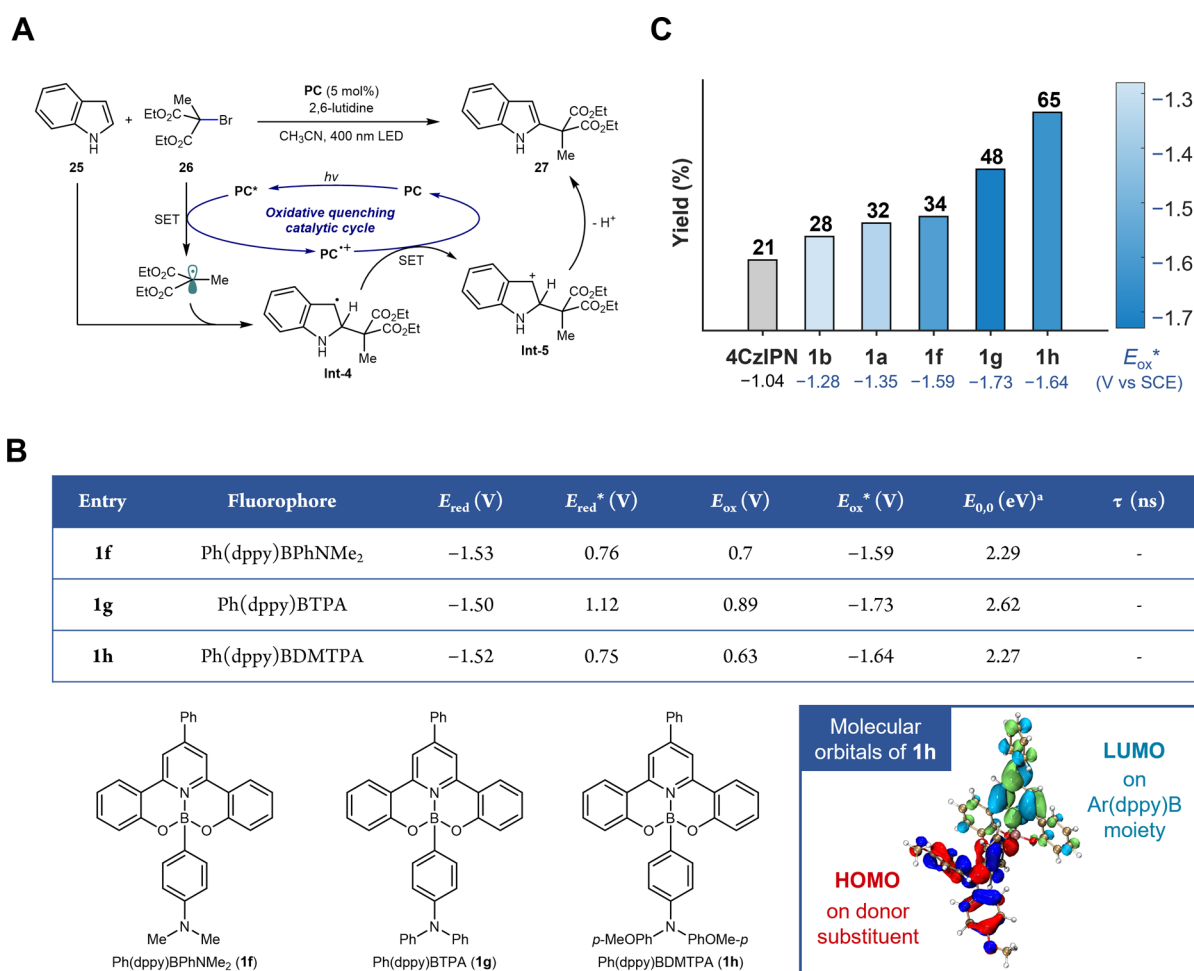


Figure 7. Structural tailoring of Ar(dppy)BX fluorophores for oxidative quenching catalysis. (A) Model reaction: a reduction-initiated addition reaction of **25** to **26**. (B) Impact of X-group modification on the excited-state oxidation potential (E_{ox}^*). Abbreviations: TPA, triphenylamino; DMTPA, dimethoxytriphenylamino. (C) Performance validation. Reaction conditions: **25** (3.0 equiv), **26** (0.5 mmol, 1.0 equiv), PC (5 mol %), 2,6-lutidine (1.5 equiv) in 1 mL of CH₃CN, 20 W LED, Ar atmosphere, rt, 8 h. Yields of **27** were determined by ¹H NMR analysis using 1,3,5-trimethoxybenzene as an internal standard.

thermodynamic tuning resulted in a marked enhancement of catalytic activity, with the yield of product **24** rising from 6% (**1e**) to a peak of 65% (**1d**), which outcompete the performance of commonly used 4CzIPN (Figure 6C). The correlation between E_{red}^* and yield in this reaction underscores the value of tuning the oxidative power of the PC* through Ar-group engineering.

4.5. Optimization for Oxidative Quenching Catalysis

Electronic tuning proved equally viable for optimizing catalytic performance in reduction-initiated reactions, using the addition of indole **25** to bromomalonate **26** as a model system (Figure 7A).⁶³ Operating via an oxidative quenching pathway, this reaction was expected to benefit from enhanced reductive power (E_{ox}^*) through the incorporation of electron-donating groups (EDGs).

To test this, we designed and synthesized catalysts featuring amine-based internal donors at the boron center (**1f–h**, Figure 7B). While these structural modifications successfully shifted E_{ox} values to more negative potentials as anticipated, they also led to dramatic fluorescence quenching, precluding the estimation of E_{ox}^* by experimental $E_{0,0}$ from the fluorescence emission spectra. Computational analysis revealed that the HOMO of **1h** is localized on the X-group (the phenylamine

ligand) rather than the Ar(dppy)B moiety. Consequently, we hypothesized that this quenching originates from the amine donors acting as intramolecular electron donors upon excitation, deactivating the locally excited (LE) state of the Ar(dppy)B moiety to form a charge-transfer (CT) state, which was further corroborated by our TD-DFT calculations (Supporting Information Section 5). As a result, the $E_{0,0}$ values for these derivatives were estimated exclusively via TD-DFT results. The E_{ox}^* , derived from calculated $E_{0,0}$ and experimental E_{ox} , were more negative than that of **1a**, and the performance of **1f–h** as PCs indeed superior to both catalysts **1a–b** and 4CzIPN exhibiting less negative E_{ox}^* (Figure 7C).

However, the resulting catalytic performance failed to follow the anticipated thermodynamic trend. Most notably, catalyst **1h** provided a significantly higher yield than **1g** (65% vs 48%), even though **1h** possesses a markedly less negative E_{ox}^* (−1.64 V) and thus a lower thermodynamic driving force for the reduction of bromomalonate. This discrepancy suggested that an alternative regulatory factor beyond simple energetics governs the reactivity of these PCs.

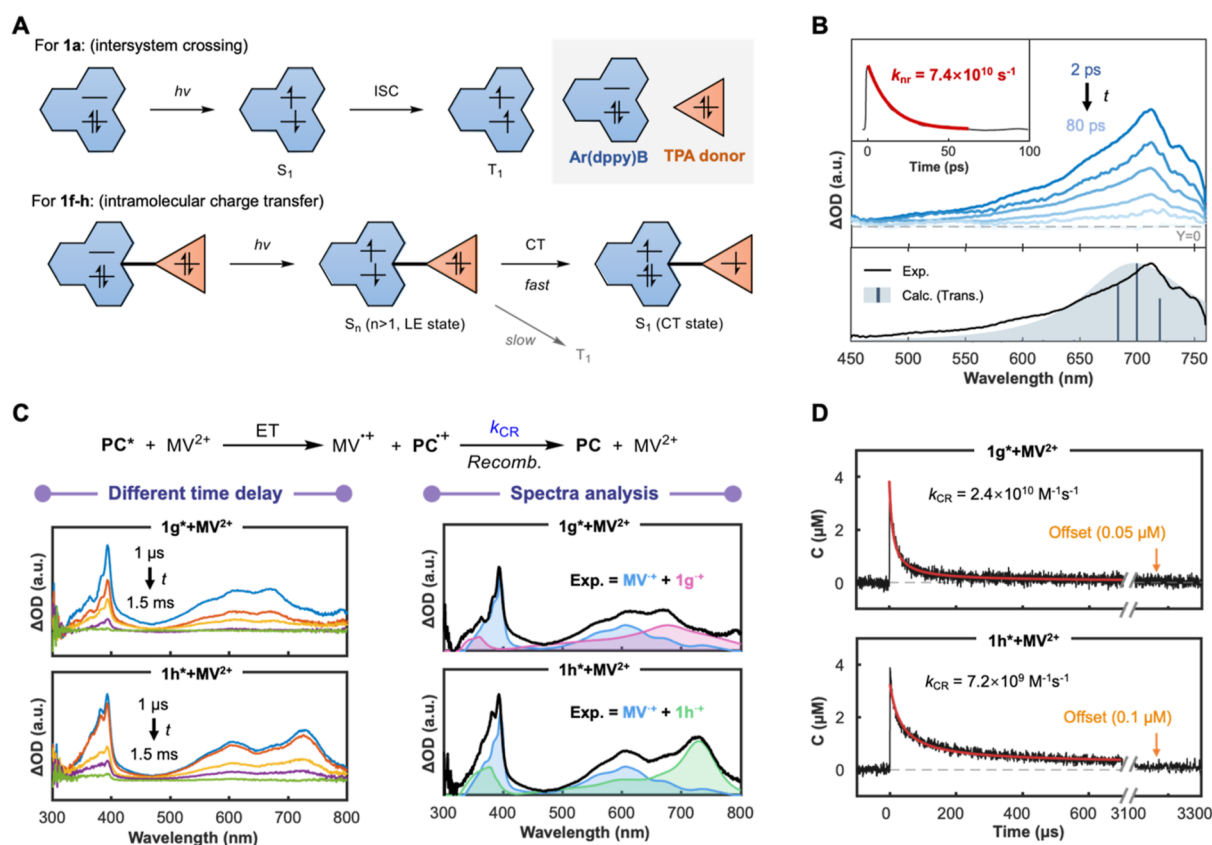


Figure 8. Mechanistic investigation of the structure–reactivity relationship in oxidative quenching cycles. (A) Proposed divergent excited-state behaviors of **1a** and **1f–h**. (B) Femtosecond transient absorption (fs-TA) spectra of **1h** showing the evolution of the CT excited state. Top inset: kinetics monitored at 710 nm. Bottom inset: comparison between experimental spectra and TD-DFT calculated results. (C) Photoinduced single-electron reduction of MV^{2+} mediated by the excited states of **1g** and **1h**. Left: TA spectra at different time delay; right: spectral assignment via superposition. (D) Comparison of $MV^{\bullet+}$ transient absorption kinetics.

4.6. Mechanistic Investigation of the Structure–Reactivity Relationship

We found that, the postexcitation behavior of **1f–h** was rather different from that of **1a** (Figure 8A). Computational analysis revealed that, in molecules **1f–h**, the $S_0 \rightarrow S_{n(n>2)}$ excitation corresponds to LE state of the pyridinium core, while $S_0 \rightarrow S_1$ excitation (with negligible oscillator strength, see Table S5) corresponds to a CT state from the donor substituent to the pyridine core. The existence of the CT state may account for the absence of fluorescence for these catalysts.

To validate the formation of this CT state, femtosecond TA (fs-TA) spectroscopy was employed (Figure 8B, top). In contrast to the fs-TA spectra of **1a'** (the more soluble bis-*tert*-butyl substituted surrogate of **1a**), which show a clear $S_1 \rightarrow T_1$ transition (Figure S17), the spectra of **1h'** (the more soluble bis-*tert*-butyl substituted surrogate of **1h**) featured a single transient species with a lifespan in sub-100 ps regime, with no other species observed. We assigned this species to the CT state of **1h'**, characterized by a nonradiative decay rate constant (k_{nr}) of $7.4 \times 10^{10} \text{ s}^{-1}$. Notably, the TD-DFT predicted absorption spectrum of the CT state for **1h** is in agreement with the experimental spectrum, further corroborating the assignment of this transient species (Figure 8B, bottom).

While such rapid decay of the CT state initially seemed to contradict the superior catalytic efficiency, monitoring the electron transfer between the excited states (**1g**^{*}/**1h**^{*}) and MV^{2+} by ns-TA indicated effective intermolecular electron

transfer (Figure 8C). For both catalysts, characteristic peaks of $MV^{\bullet+}$ and **1g**^{*}/**1h**^{*} were observed shortly after the laser pulse (1 μs), decaying nearly to baseline within 1.5 ms. Since both $MV^{\bullet+}$ and **1g**^{*}/**1h**^{*} are persistent species, this observation indicates efficient charge recombination. Kinetic analysis (Figure 8D) revealed that less than 3% of catalysts **1g** and **1h** decomposed per excitation cycle, a significant improvement over the 15% degradation observed for **1a**. Furthermore, **1h**^{*} exhibited a slower charge recombination rate ($k_{CR} = 7.2 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$) compared to **1g**^{*} ($k_{CR} = 2.4 \times 10^{10} \text{ M}^{-1}\text{s}^{-1}$), likely due to favorable potential differences and electronic stabilization afforded by the DMTPA motif.

These findings demonstrate that while E_{ox}^* defines the thermodynamic feasibility of the reduction step, the inhibition of catalyst degradation and the kinetic suppression of unproductive charge recombination via structural refinement might be the primary determinants of catalytic throughput in oxidative quenching cycles. This underscores the versatility of the Ar(dppy)BX photocatalyst manifold, which facilitates diverse reactions through the synergistic tailoring of redox potentials, catalyst stability, and charge recombination kinetics.

5. CONCLUSIONS

In summary, we have demonstrated the potential of pyridinium(boron) fluorophores, specifically the Ar(dppy)BX motif, as versatile photocatalysts for a range of chemical transformations. Our investigation revealed that this photocatalyst exhibits excellent photoredox and energy transfer

capabilities, rivaling the performance of established photocatalysts such as 4CzIPN.

The ability to selectively control reaction pathways, such as in the photocatalyzed switchable oxidation of amines, showcases the practical utility of this type of photocatalyst in synthetic organic chemistry. By tuning the structure of the substituents on the fluorophore, we achieved modifiable redox properties, enabling targeted reactions across varied substrates. The efficacy of the Ar(dppy)BX photocatalyst was exemplified through its application in both oxidation-initiated and reduction-initiated reactions, where we observed significant enhancements in product yields correlated with tuning of oxidation or reduction potentials. These findings suggest that the Ar(dppy)BX family of fluorophores holds great promise for advanced photocatalytic applications.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.6c04010>.

The single-crystal X-ray diffraction data for structures reported herein have been deposited at the Cambridge Crystallographic Data Centre (CCDC), under deposition numbers Details for experimental procedures, instrument setup, and compound characterization (PDF)

Accession Codes

Deposition Numbers 2340929, 2340935, and 2341012 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe [Access Structures service](#).

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Notes

The authors declare no competing financial interest.

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